

Update: October 2025

In October 2025 [MAK](#) published their [final report](#) (in Russian) on this accident. This added great deal of data to our knowledge of the dynamics of this accident.

I have added a section below comparing that data with my model that predates the investigation conclusion.

AN-24 Accident at Nizhneangarsk, Russia

On Thursday 27 June 2019 an AN-24 of Angara Airlines, flight number 2G-200, flying from Ulan-Ude Airport was destroyed whilst landing at Nizhneangarsk Airport. Reportedly, the left engine had failed in flight.

A video of the landing was posted online apparently taken by a passenger sitting on the right hand side under the wing (this record is called here video 'A'). A second video (video 'B') emerged that was taken from a fixed security camera overlooking the airfield. In both videos the aircraft approaches at high speed and after touchdown it veers off the runway to the right eventually colliding with a series of obstructions.

This is an analysis of these videos that gives a picture of the speed, position and acceleration forces experienced by the aircraft.

Summary of Observations

Times are video 'A' unless stated otherwise.

- At the start of the video A the aircraft is on the extended centreline of Nizhneangarsk runway 23 (length 1653 m) at a distance of 2482 ± 98 m from the start of the runway. Ground speed is 82.8 ± 2.0 m/s, (161 ± 4 knots).
- The ground speed slowly increases to a maximum of 92.7 ± 3.5 m/s (180 ± 7 knots) at around $t=17.0$ seconds when the aircraft is 964 ± 44 m from runway 23.
- The aircraft crosses the start of runway 23 at $t=27.5$ travelling at 88.6 ± 2 m/s (172 ± 4 knots).
- Touchdown is at $t=33.8$ s, 549 ± 14 m from the runway start (about 1/3rd of the way down) at 84.3 ± 2 m/s (164 ± 4 knots). Acceleration is -1.1 m/s².
- At $t=36.0$ the aircraft starts drifting to the right, possibly due to a burst tyre. The aircraft is 731 ± 16 m down the runway travelling at 81.8 ± 2 m/s (159 ± 4 knots).
- The aircraft departs the runway at $t=46.1$ s, 1463 ± 41 m down the runway at 58.7 ± 3.0 m/s (114 ± 6 knots). Acceleration is -3.9 m/s².
- At $t=56.1$ there is an impact with the boundary fence 1853m from the start of the runway. This collapses the starboard undercarriage (and probably the port one as well). Impact speed is around 19 m/s (37 knots).
- The aircraft would have slid to a stop roughly 45m later unfortunately there was a building 36m from the boundary fence which is the site of the final impact. Time is about $t=58.0$ and speed is somewhat less than 9 m/s (18 knots).

The annotated image below shows the estimated ground path of the aircraft in dotted red with significant events in yellow:



Resources

Apart from the videos, there is:

- Wikipedia on [Angara Airlines Flight 200](#) and on [Nizhneangarsk Airport](#)
- [AV Herald](#)
- [Aviation Safety Network](#)
- Videos from [RT](#)
- Hi resolution ($\sim 0.6m$ per pixel) imagery of the area, for example from [google maps](#).

Data From Video A

Video A <https://youtu.be/LtJcgdU5MUk> is taken from inside the aircraft.



The following techniques are used to calculate the position of the aircraft during this video:

- Matching landmarks in the video with aerial imagery.
- Examining runway details in the video to estimate the aircrafts ground speed.
- Calculating the likely progress of a ballistic body.

Video metadata shows the video was shot at 30 frames per second.

Landmarks in the Video

The video shows some distinct landmarks, in particular some colourful and readily identifiable buildings which can be seen on aerial imagery. These are used to find the aircraft position on the extended centreline of runway 23 (the aircraft appears to make a straight in approach with little lateral deviation).

Here are two examples. Frame 1 ($t=0s$) shows the line up of the edge of a settlement with a dark smudge on a sandy part of an island. The magenta line shows the estimated ground line the aircraft is flying across, the red line is the extended centreline of runway 23:



This puts the aircraft $2490\pm98m$ from the start of runway 23.

The second example is frame 483 ($t=16.0s$) where a two distinctive buildings are lined up:

Camera View**Aerial View**

This puts the aircraft at $1041 \pm 49\text{m}$ from the start of runway 23.

There are numerous opportunities here but around 15 measurements are sufficient to establish the flight path. This measurement is of distance and this can be differentiated to give velocity. Errors in this method is estimated to be $\pm 100\text{m}$ when 2500m from the runway, reducing to $\pm 10\text{m}$ at the runway edge.

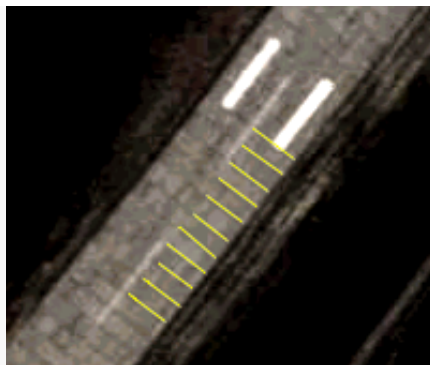
This technique is useful but it requires identifiable ground features and few exist after the aircraft crosses the threshold, and none once 330m beyond the start of runway 23 (the runway markings are useful). Fortunately another technique exists due to the particular construction of this runway which is typical of Siberian airports. Even better, these two (independent) techniques overlap for about four seconds so that the analysis can be cross checked.

Counting Slabs

The runway at Nizhneangarsk is made of concrete slabs that are ubiquitous in Siberia for roads and airfields. This gives the runway a particular pattern where vegetation or water staining at the slab edges contrasts with the pale concrete of the slab. These edges can be clearly seen, and measured, on the video and high-resolution aerial imagery. The aircraft's progress across these slabs can be used to calculate the velocity. This can be integrated for distance or differentiated for acceleration.

Standard Slabs

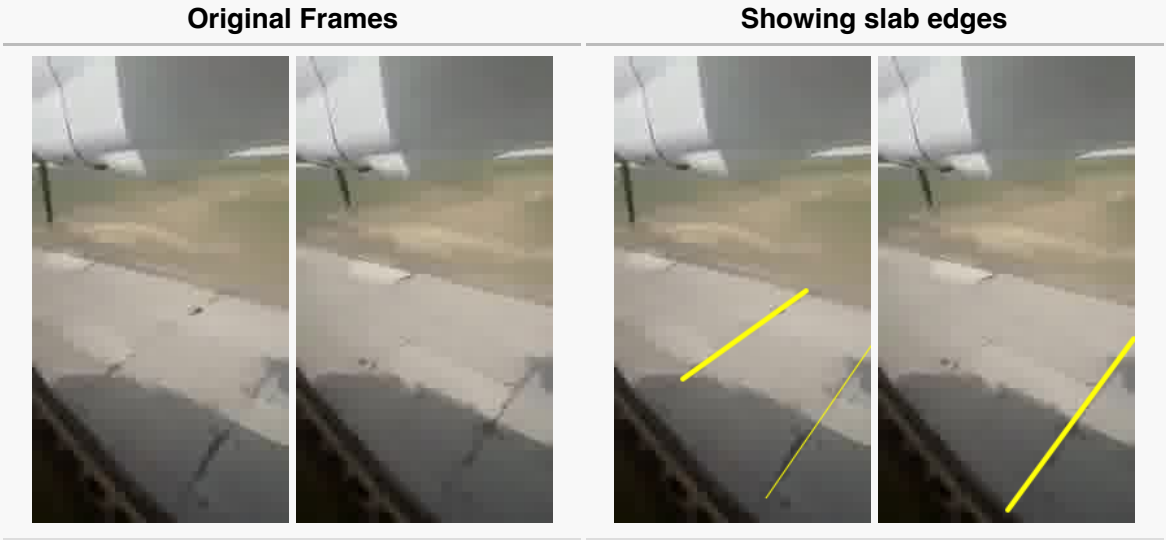
Standardised slabs used for roads and runways in Siberia measure $6\text{m} \times 1.8\text{m}$ (data from the author's engineering file). For confirmation here is an enhanced aerial image of the runway at Nizhneangarsk where the south side of the runway has been annotated, rather crudely, with yellow lines at the edge of each slab (the slabs are laid with their long side in the runway direction).



Measurement of this imagery shows that these 10 slabs are 98 pixels long which, at 0.617 m/pixel for this image, gives a slab length of 6.05m. The value of 6m for slab length is used hereon.

Speed by Slab Transits

These slab edges can be seen quite clearly on video 'A', as an example here are two frames [841 and 843] which are 1/15 second apart (at 30 fps). On the left are the original frames, on the right the edges of the slabs are identified with yellow lines.



The broad yellow line on the first image is the same part of the runway as the broad line on the second image. The near match of the right hand line in the first frame with the line in the second frame indicates that the aircraft has traveled almost exactly 6m in that 1/15th second. The light patch on the right of the concrete, and the lightish patch in the grass confirms this sequence.

This gives a ground speed of 90 m/s (175 knots). The error term is how confident the slab match is, in general this assumed to be 10% of a slab, so 0.6m in this example thus ± 9 m/s (± 18 knots). If a succession of slabs can be counted the error is proportionally reduced. This is a speed measurement and can be integrated for distance or differentiated for acceleration.

Data From Video B

A second video <https://youtu.be/BQ8ujmRhLH0> was posted on YouTube that showed images from a security camera overlooking the airport, here referred to as video 'B'. The camera is facing roughly south west and was recorded at 25 frames per second.



Camera Position

The distinctive foreground means the camera position is fairly easily identified on aerial imagery from Google Earth at a position $x=491\text{m}$ down runway 23 and $y=-161\text{m}$ to the left of the runway centreline.

Camera Heading and Resolution

The first calculation is to find the camera heading (degrees) and the resolution (pixels per degree). To do this various static objects that are identifiable on aerial imagery are used:

- A Mi-2 helicopter near the centre of the frame. The length and height of this can also be used to help calibrate the camera.
- Towards the right of the frame is a pale smudge where tarmac meets grass.
- Slightly left of centre is a bright roofed house beyond the tree-line.
- There are some structures on the extreme left of the frame.
- The red building identified as the scene of the final impact can be determined by the base of the column of smoke that starts at around frame 850.

Comparing the relative frame positions of these with the actual bearings gives a camera resolution of 21.6 ± 1.9 (pixels/degree), this assumes a linear field. For a video that is 640 pixels wide this corresponds to an equivalent focal length for a 35mm camera of $68 \pm 6\text{mm}$. This also means that the centreline of the camera is pointing at a bearing of 238.7 ± 2.1 degrees.

Matching TimeLines

Several events can be seen in video B that can be compared with video A and this correlation can be used to match the timelines of the two videos:

Video B Event	Video B Time (s)	Video A Event	Video A Time (s)	Difference (s)
First appearance, bearing 253.5° from camera	1.4	First appearance in video B, x=688±19 m	35.5	34.1
Tyre smoke and dust	1.6	Start of drift to the right.	36.0	34.4
Start of large dust plume	11.6	Runway disappears	46.1	34.5
Start of smoke plume	~23	Final impact?	~57	34

Table 1: Matching timelines of video A and B.

This gives a mean time difference of video A and B of 34.25±0.3 (s). The consistent match over 21 seconds show that the two videos are on the same timeline.

Computed Results

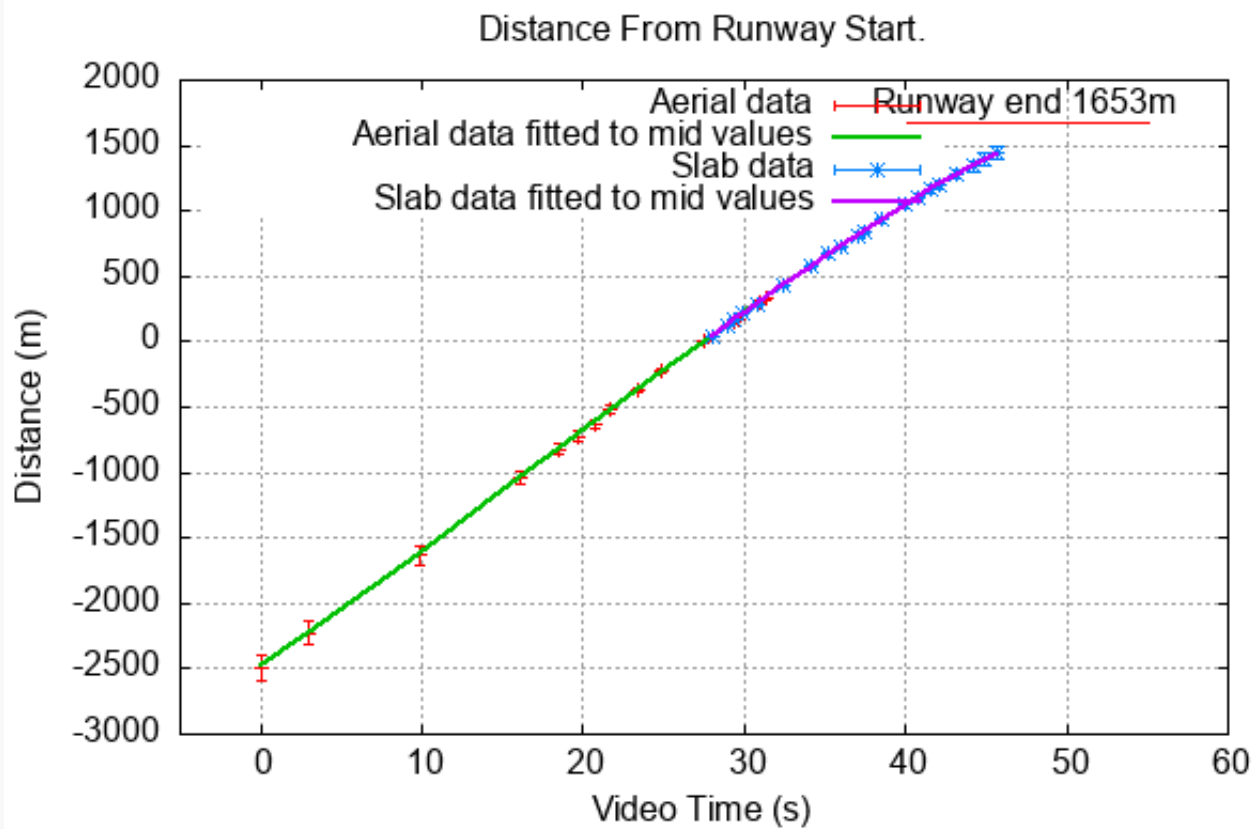
Video A

Aircraft Position and Speed

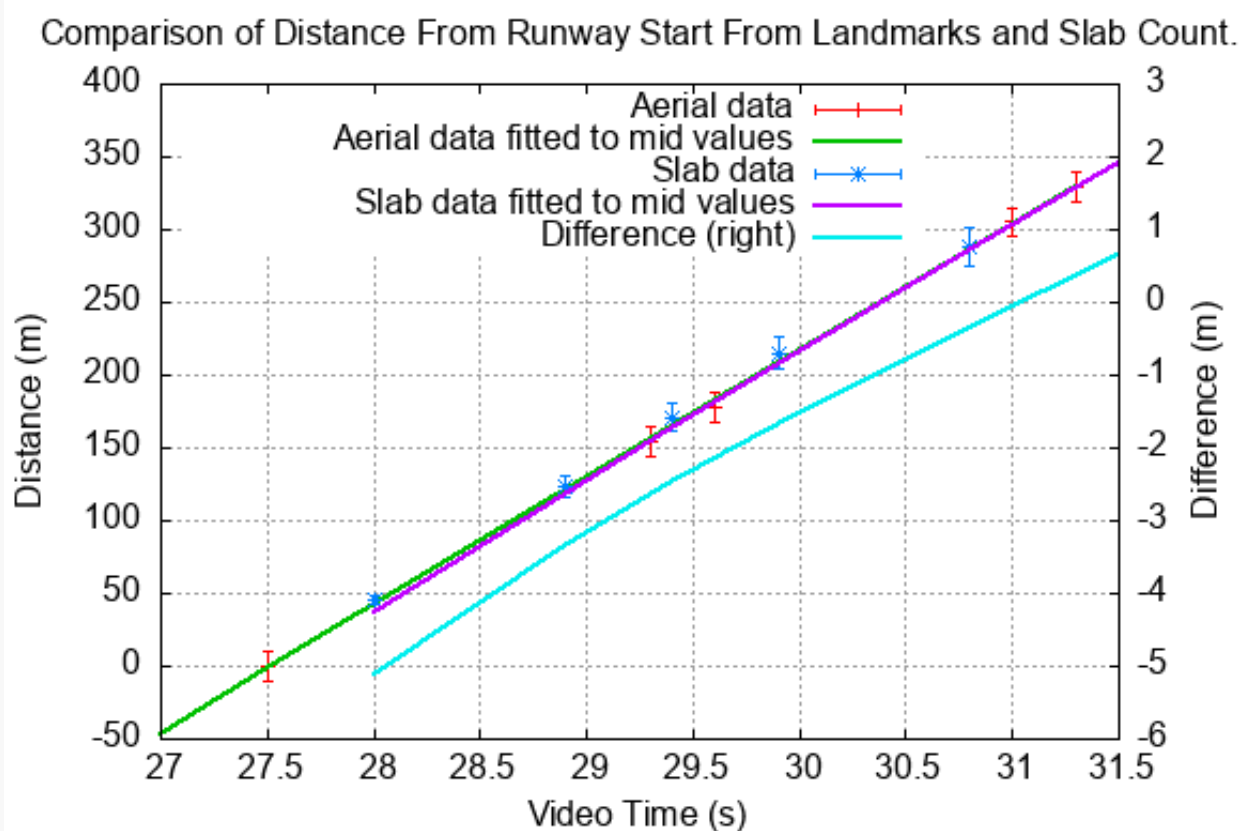
Analysis of video A gives:

- The aircraft position from overflying landmarks, differentiating this give the aircraft's ground speed.
- The aircraft's passage over the runway slabs gives the aircrafts ground speed, integrating this gives the distance that the aircraft has covered.

Here is a plot of distance from the start of runway 23 over time from video A, the mid points are fitted to a third order polynomial. The right hand graph shows the comparison where the two measurements overlap:



The errors are:



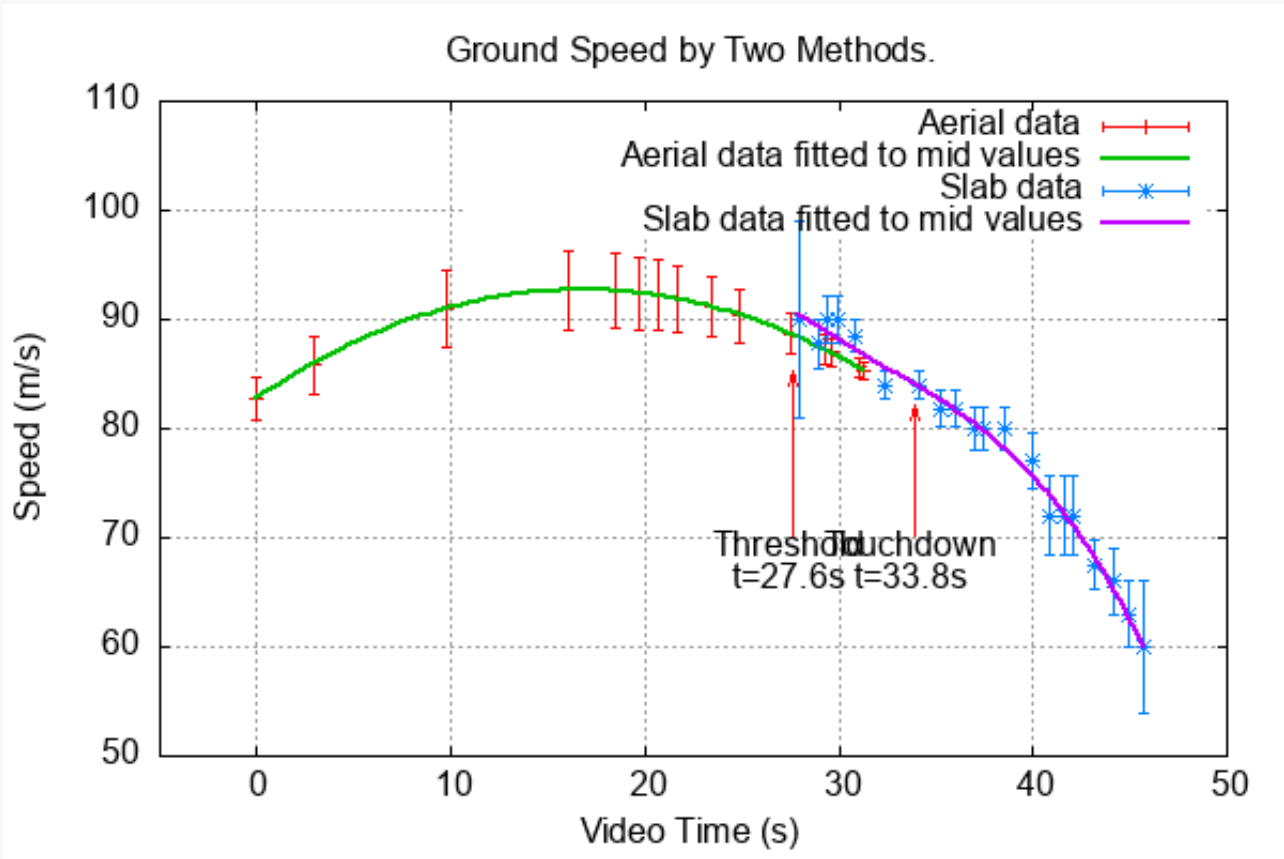
The difference between the two techniques is 5m or less.

Aircraft Ground Speed

The aircraft's ground speed can be calculated from video A by:

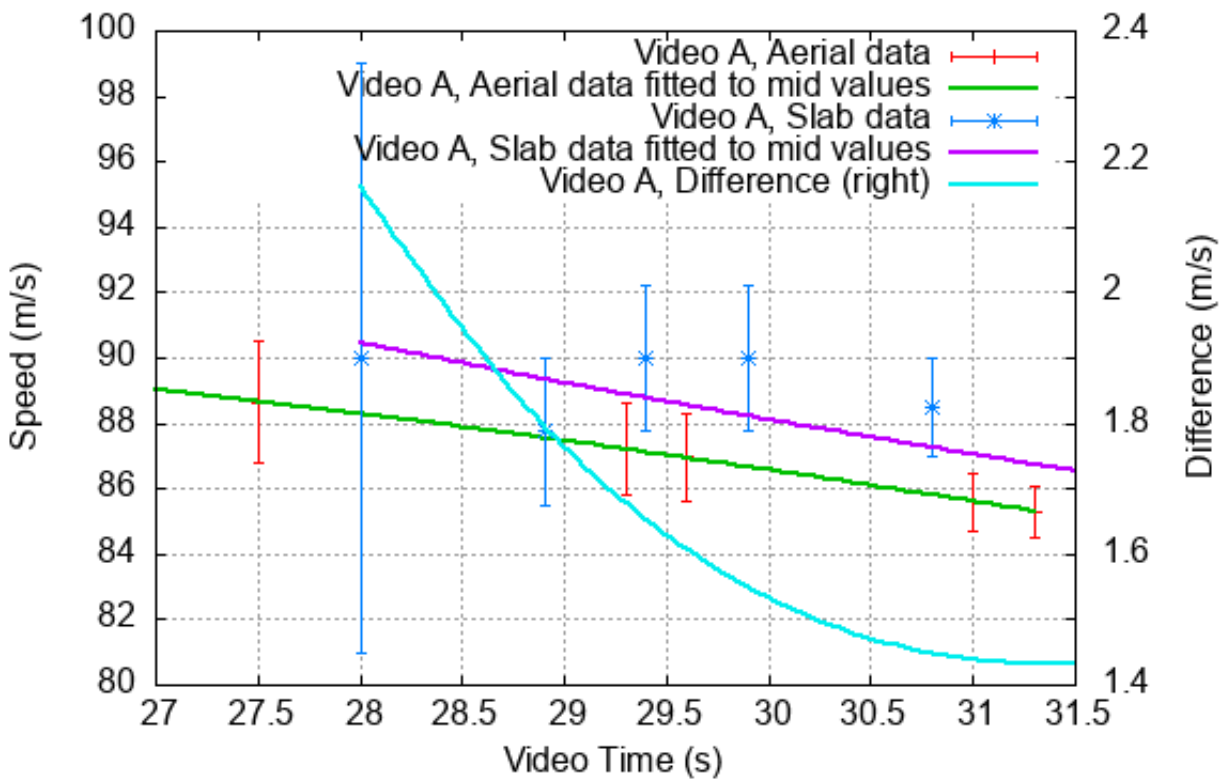
- The differential of the aircraft's position when overflying landmarks.
- Directly from the aircraft's rate of passage over the runway slabs.

Here is the ground speed by the two techniques. The right hand graph shows the comparison where the two measurements overlap:



The errors are:

Comparison of Ground Speed From Landmarks and Slab Count.



The difference is 1.4 to 2.2 m/s over this interval with the fitted data which is just about within the error terms of each if centred on the respective fitted data.

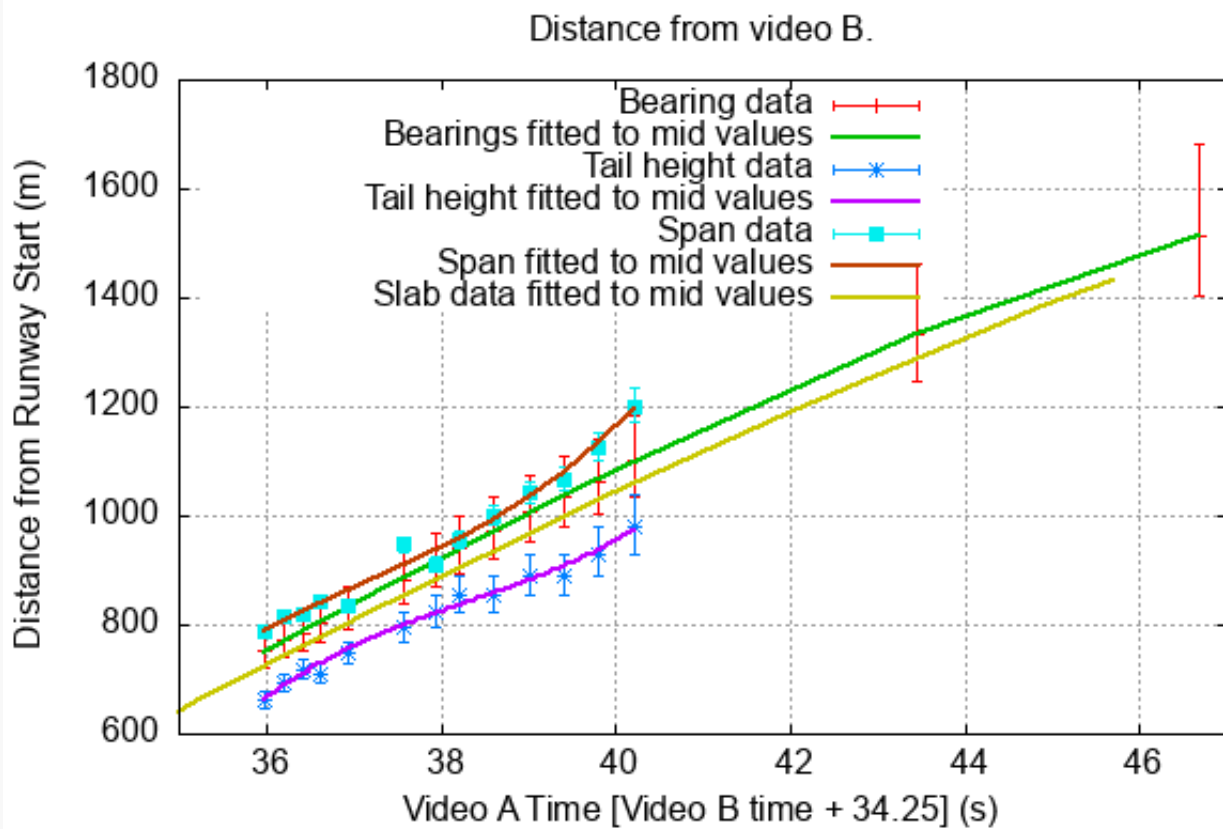
Video B

Various unsuccessful attempts were made to identify the aircraft position and speed from video B (the security camera video). The following measurements, once the camera heading and resolution were known, could be converted to the aircraft's position:

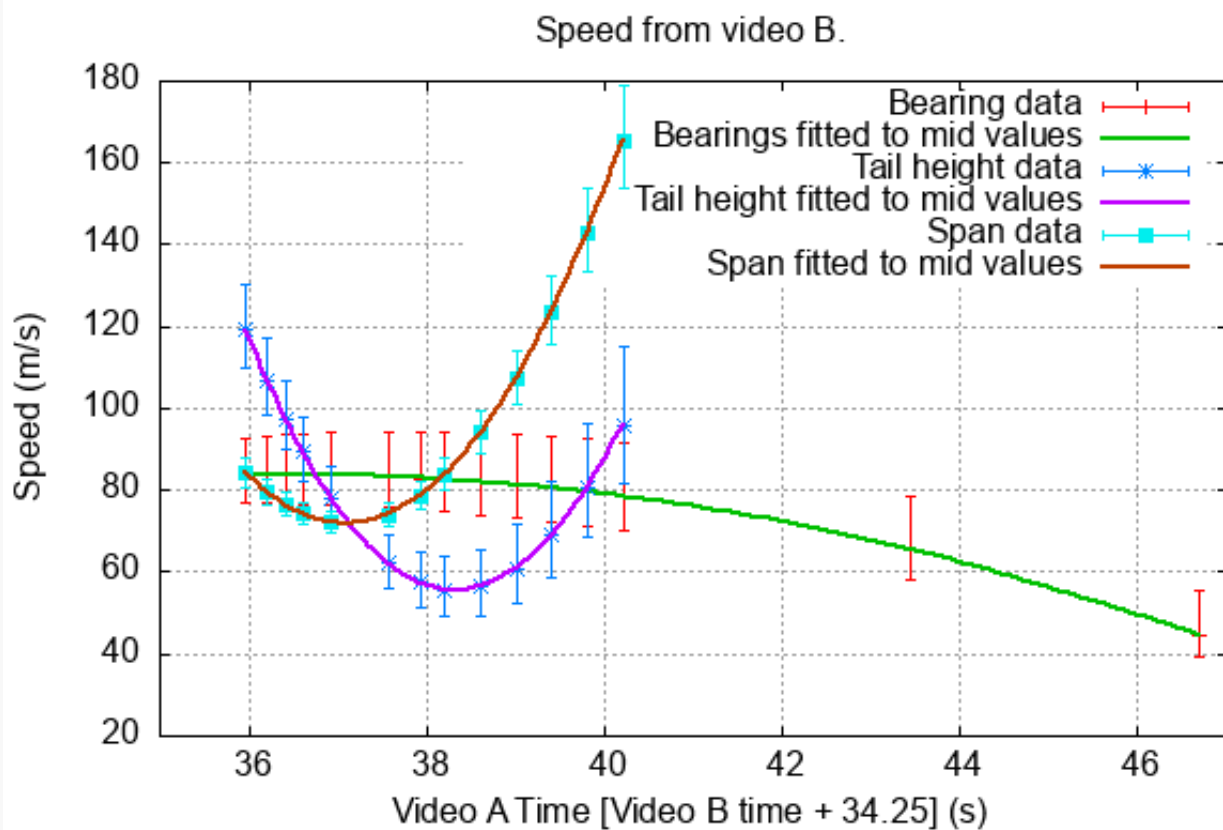
1. The bearing of the aircraft from the camera.
2. The range of the aircraft from the apparent height of the top of the tail above ground.
3. The range of the aircraft from the apparent span of the wings.

A problem for the latter two is that the measurements are often only a few pixels wide with a correspondingly large uncertainty. Beyond $t=40.2$ the data from the latter two techniques is deemed unreliable due to the low resolution of the camera and the indistinct image of the aircraft.

The calculated distance down the runway by these three methods agree rather roughly in magnitude as shown below on the left (the distance data from the slab measurements is also shown for comparison). However when differentiated to obtain ground speed the results are implausible except for the bearing data, as seen on the right.

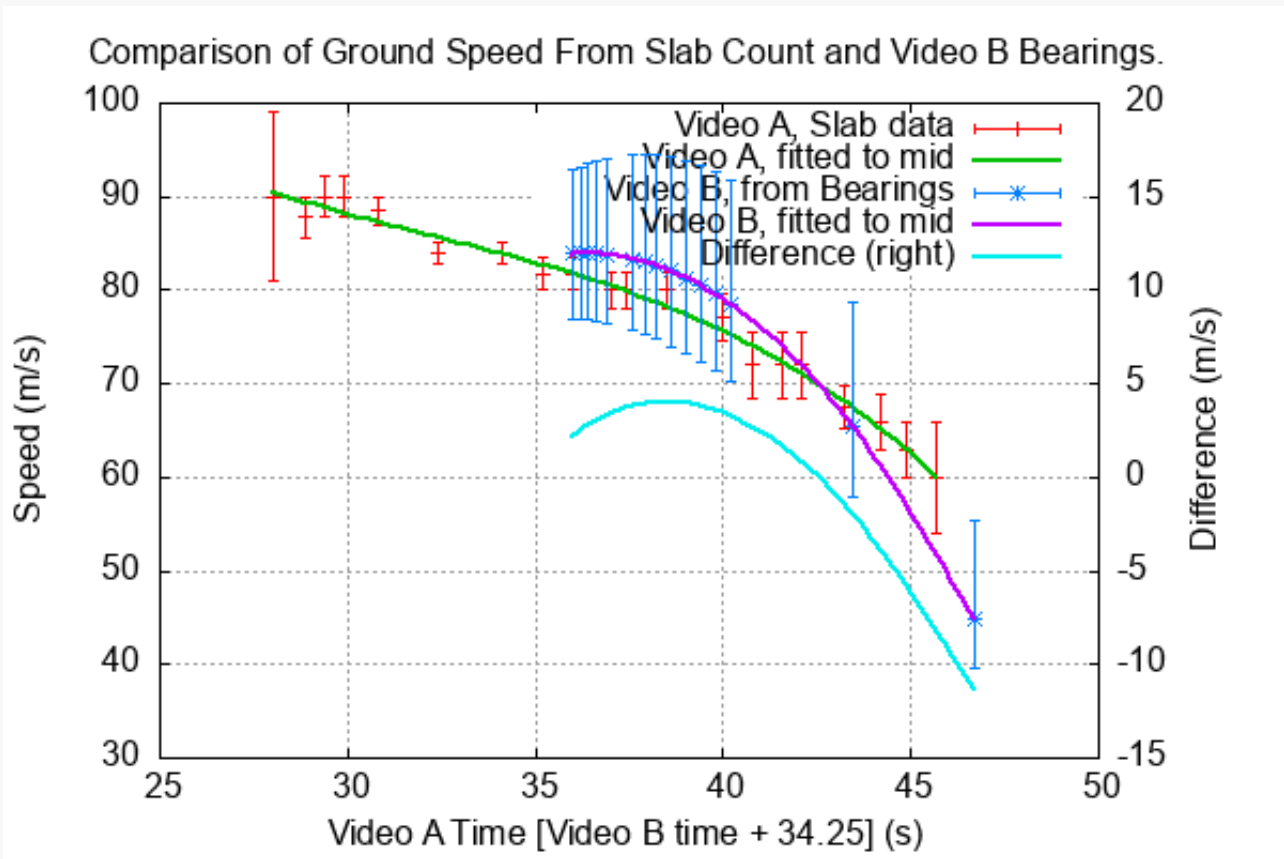


The errors are:



The poor results from this analysis could be for many reasons; the low resolution of the video, measurement error, non-linear camera field and so on. Even the bearing data does not make sense giving a de-acceleration of around -5m/s, if this were true

the aircraft would have stopped well before the boundary fence which was not the case. The following graph compares the ground speed from the slab speed data of video A to the bearing data of video B:



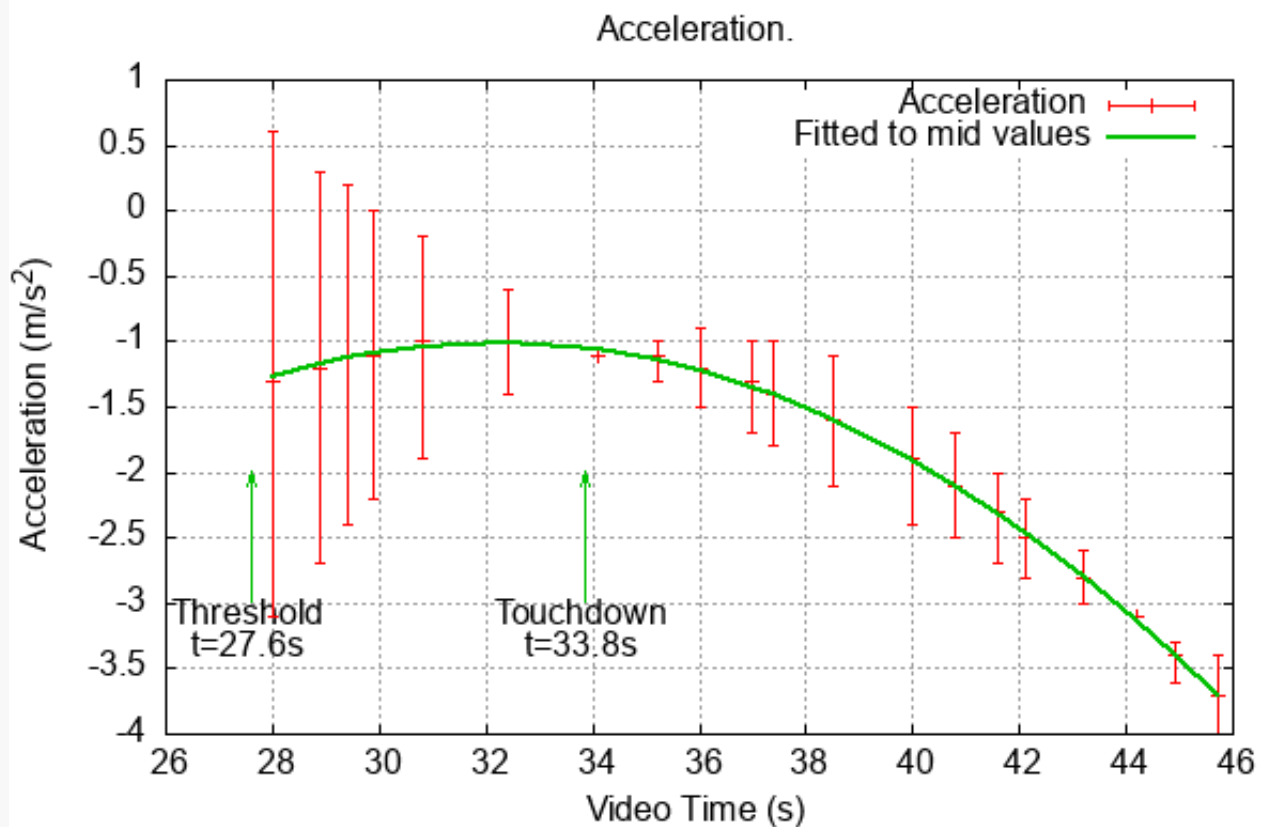
Conclusion for the Data From Video B

The data obtained from video B is not useful for calculating the position or speed of the aircraft. The data from video B is useful in that it can tie visible events to the motion of the aircraft seen in video A, for example the smoke and dust appearing from the tyres and the start of the aircraft swinging to the right.

The video A data is used henceforth to compute the trajectory of the aircraft.

Acceleration

The ground speed if the aircraft computed from video A can be differentiated to give the longitudinal acceleration:



There are no visible discontinuities of sudden changes of acceleration such as might be caused by a tyre burst or running of the runway. Possibly this is because of the smoothing caused by the curve fitting process.

Motion After t=45.7 Seconds

The last computed position at t=45.7 seconds is:

Time	Position	Ground Speed	Acceleration
45.7 s	1442 ±48 m	60.0 ±3.7 m/s, 117 ±12 knots	-3.7 ±0.3 m/s ²

Table 2: Last computed position.

There are no landmarks or visible features in video A after t=45.7 seconds that can be used to give speed or position but there is a collision with the boundary fence that can be used to estimate these values.

The Boundary Fence Impact

At t=56.1s the aircraft collides with the boundary fence and this fence is readily identifiable on the aerial imagery as being 1853m from the start of runway 23, 200m beyond the runway end. The sequence below shows four consecutive frames from the video of the starboard undercarriage collapse as the aircraft goes through the airfield boundary.



The first frame [1684] shows the moment before impact, the obstruction is on the left which looks like a substantial concrete block, the dark disc at the bottom is the tyre and the silver vertical is the undercarriage leg. The second frame ($t=56.1\text{s}$) shows the moment of impact and the last two frames show progressive collapse of the starboard landing leg. The last frame is rotated somewhat to the right, most likely due to the forces on the person filming which would naturally rotate a hand held camera to the left.

Here are those frames as a composite image adjusting, approximately, for alignment and tilt:



Boundary Fence Impact Speed and Acceleration

The impact point is 1853m from the start of runway 23 and 10.4 seconds after the last known position. Given the initial speed and initial position the mean speed can be calculated. Assuming constant acceleration the impact speed and the acceleration can be calculated. For example, given v initial as 60.0 m/s at 1442 m:

- Then v mean is distance over time; $(1853 - 1442) / 10.4 = 39.5 \text{ m/s}$.
- So v final is $39.5 - (60.0 - 39.5) = 19.1 \text{ m/s}$
- Giving an average acceleration of $(19.0 - 60.0) / 10.4 = -3.9 \text{ m/s}^2$.

Here is that calculation for the mid point and the +/- error estimates.

Calculation	v initial in m/s (knots)	d initial (m)	a (m/s ²)	v impact in m/s (knots)
Mid point	60.0 (117)	1442	-3.9	19.1 (37)
+ve error	63.8 (124)	1489	-5.5	6.2 (12)
-ve error	56.3 (109)	1394	-2.3	32.0 (62)

Table 3: Estimated boundary fence impact speed.

The mid point data computed acceleration of -3.9 m/s² agrees very well with the last computed acceleration of -3.7 ±0.3 m/s². That the mid estimate is correct is further supported by the ballistic (i.e. uncontrolled) trajectory of the aircraft to the final impact.

Final Impact

36m beyond the boundary fence is a building that is where the aircraft finally stopped. Assuming the (de)acceleration is the same as before the boundary fence impact and ignoring any speed reduction caused by the boundary fence impact the final impact speed would be:

Calculation	v boundary fence, from table 3 in m/s (knots)	a, from table 3 (m/s ²)	Calculated distance to full stop (m)	v impact at 36m (m/s, knots)
Mid point	19.1 (37)	-3.9	46.4	9.1 (18)
+ve error	6.2 (12)	-5.5	3.5	No impact
-ve error	32.0 (62)	-2.3	219	29.3 (57)

Table 4: Estimated final impact speed.

Images of the aftermath suggest that the mid-point calculation is probably correct and that the final impact occurred at roughly 9 m/s (18 knots). This can be regarded as an upper bound as it ignores the speed lost by the collision with the boundary fence and the likely increase of deceleration after the undercarriage collapsed.

Summary of Events

Combining all the data gives the following summary of events, time is in video A time, position is from the start of runway 23:

Time (s)	Position (m)	Ground Speed (m/s, knots)	Acceleration (m/s ²)	Description
0.0	-2481±98	82.8±2.0, 161±4	1.2±0.2	Video start.
17.0	-964±44	92.7±3.5, 180±7	0.0±0.1	Maximum ground speed.
27.6	0±14	88.6±1.9, 172±4	-0.7±0.1	Threshold.
33.8	549±17	84.3±0.8, 164±2	-1.0±0.1	Touchdown.
35.5	688±19	82.4±1.0, 160±2	-1.2±0.2	First appearance in video B.
36.0	731±19	81.8±1.1, 159±2	-1.2±0.3	Start of drift to the right. Tyre smoke and dust observed in video B.
45.7	1442±48	60.0±3.7, 117±7	-3.7±0.3	Last speed measurement.
46.1	1463±49	~58.7±3.6, 114±7	-3.9±0.4	Runway disappears. Dust plume observed in video B. Speed and distance are extrapolated.
56.1	1853±10	Around: 19, 37	Around -3.9	Impact with boundary fence.
~57	1889±10	Less than: 9, 18		Final impact. Smoke column starts in video B.

Table 5: Selected Events

Given that the aircraft crosses the threshold at video time of 27.6 then these are the time relative to that.

Time (s)	Position (m)	Ground Speed (m/s, knots)	Acceleration (m/s ²)	Description
-27.6	-2481±98	82.8±2.0, 161±4	1.2±0.2	Video start.
-8.6	-964±44	92.7±3.5, 180±7	0.0±0.1	Maximum ground speed.
0.0	0±14	88.6±1.9, 172±4	-0.7±0.1	Threshold.
6.2	549±17	84.3±0.8, 164±2	-1.0±0.1	Touchdown.
7.9	688±19	82.4±1.0, 160±2	-1.2±0.2	First appearance in video B.
8.4	731±19	81.8±1.1, 159±2	-1.2±0.3	Start of drift to the right. Tyre smoke and dust observed in video B.
18.1	1442±48	60.0±3.7, 117±7	-3.7±0.3	Last speed measurement.
18.5	1463±49	~58.7±3.6, 114±7	-3.9±0.4	Runway disappears. Dust plume observed in video B. Speed and distance are extrapolated.
28.5	1853±10	Around: 19, 37	Around -3.9	Impact with boundary fence.
~29.4	1889±10	Less than: 9, 18		Final impact. Smoke column starts in video B.

Table 6: Selected Events with Datum Time at the Runway Threshold

Note on Sources and Accuracy

Video 'A' is the main source of data. There are two data sources from that video:

- The 'tile' data from that video is in time range: -27.6 to 3.7 (s). That data is based on locating the aircraft position and working out its ground-speed from that data.
- The 'slab' data from that video is in time range: 0.467 to 18.1 (s). That data is based on measuring the aircraft speed and working out its position by integrating that data.

Several points:

- The two data sources overlap in the range 0.467 to 3.733 (s) and this can be used to cross-check the computation (as above).
- Any position prior to -27.6 (s) is an extrapolation.
- Any position after to 18.1 (s) is an extrapolation. Since integration is involved then errors will increase markedly over time.

Sources of information Subsequently Discovered

These sources of information were discovered after this initial investigation ended on 2019-08-26:

- The Russian Accident Investigation: <https://mak-iac.org/en/rassledovaniya/an-24rv-ra-47366-27-06-2019/>
- Preliminary Accident Report (in Russian) Created 1 Aug 2019 at 10:16, modified 11 Oct 2019 at 13:30: <https://mak-iac.org/upload/iblock/870/reportra-47366pr.pdf>

Comparison with the Final Report

This final report was published in October 2025 on the [MAK website for RA-47366](#) as the [final report](#).

From the final report, page 9, "The investigation began on June 27, 2019. The investigation was completed on April 13, 2022". The final report metadata shows that the final report modification date is "8 Oct 2025 at 11:23".

For this analysis the time datum is the crossing of the runway threshold. This happens at frame 827 of Video A, 27.57 seconds after the start of that video. According to the final report this happened at 02:24:20.30 UTC.

Comparison of Significant Events

Event	Final (s)	Me (s)	Diff (s)	Final (m)	Me (m)	Diff (m)	Notes
Touchdown	7.8	6.2	-1.6	661	549	-112	See "Touchdown Time" below.
Depart Runway	18.2	18.5	0.3	1494	1463	-31.4	Not a precise event.
Boundary Wall	N/A	28.5	N/A	1844	1853	8.8	
Final Impact	28.3	29.4	1.1	1898	1889	-9.3	

Touchdown Time

I can't explain this discrepancy. To me, Video A is clear with the jolt to the passenger in frame 1015.

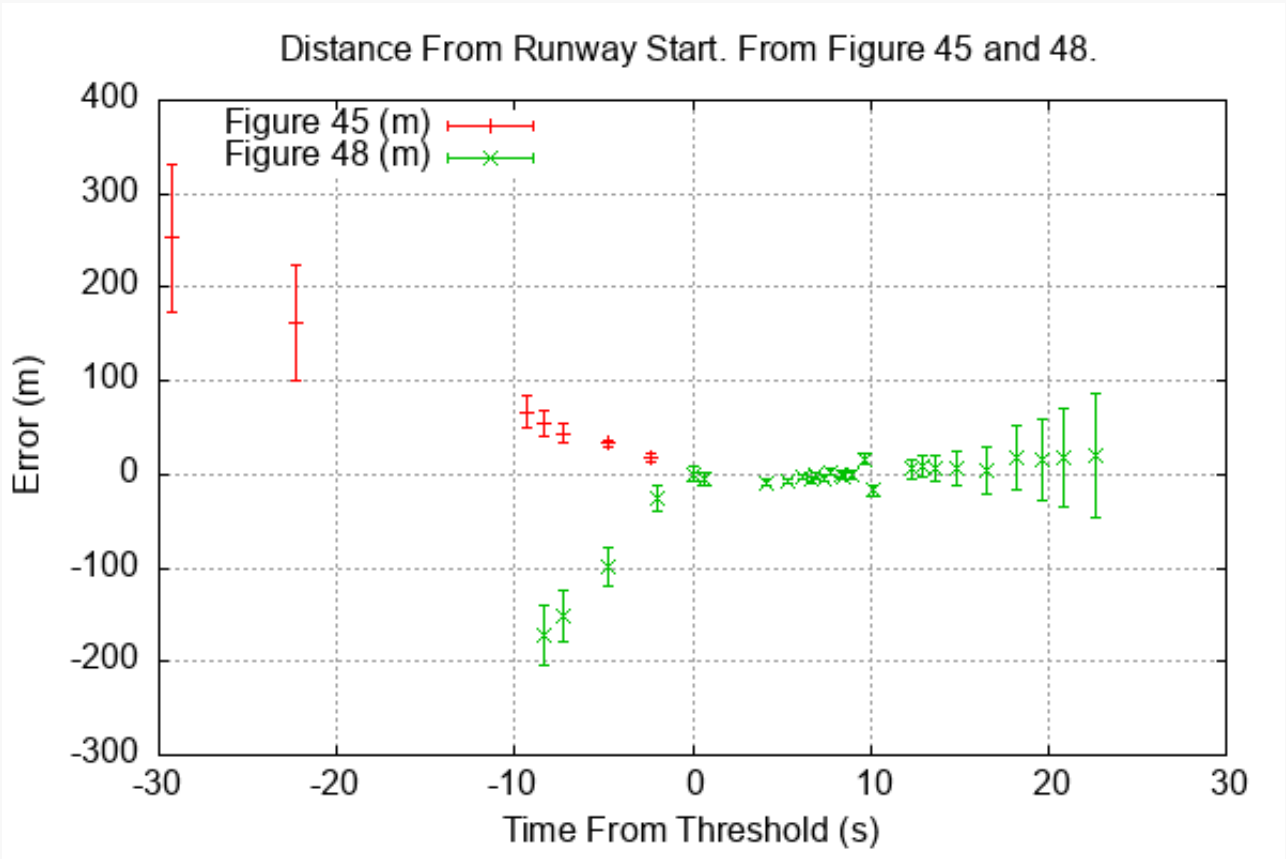
Another anomaly is that in the Flight Data Recorder data presented in Figure 46 on page 111 of the final report there is a clear 'g[y]' event (which continues) starting at +12 (s) after the datum. It is the purple curve. Is this a record of the touchdown?

Comparing the Aircraft Position with my Model

In the report are two figures of interest that plot the aircraft's position over the ground. On page 110 is Figure 45 that shows the initial approach. On page 116 is Figure 48 that shows the landing. The aircraft movement described in these figures can be compared with my calculations.

Here is the comparison, on the x-axis is time relative to the threshold crossing time. The y-axis shows the difference between my calculations and the positions shown in the two figures.

My model is only valid from -27.6 to 18.1 (s) but can be extrapolated with some confidence.



What is clear from this is that the

- Prior to touchdown there is significant divergence from my model and the final report.
- After touchdown my model agrees very closely with the final report. The increased error range over time is a natural consequence of integrating velocity data to determine position.
- There is an unexplained anomaly in the final report; the two figures 45 and 48 do not agree with their plots of the aircraft positions in five places in the 10 seconds before touchdown. I checked these carefully with two different methods and obtained substantially the same results so I suspect that this is a minor plotting error in the final report. This is why there is a divergence between the green and red graphs in the range -10 to 0 (s) as we are comparing two different values against a single one (mine).

Otherwise there is a satisfying correlation between my data and the final report. In particular the use of 'slab' data to extrapolate the aircraft's position worked far better than I dared hope.

This work was completed in July and August 2019, well ahead of the investigation which was completed on April 13, 2022.